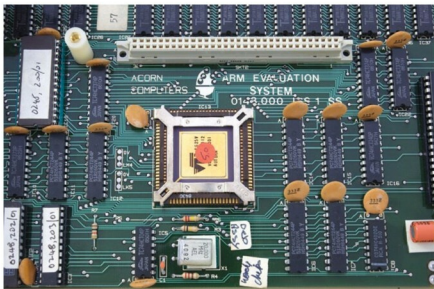


Introduction to ARM Processor

ARM is a family of instruction set architectures used in making computer processors developed by ARM Holdings. It is based on the reduced instruction set computing architecture which is commonly called as RISC. In 2010, ARM Holdings, plc reported shipments of approximately 6100 million ARM-based processors to manufacturers of chips based on ARM architectures, representing 35 per cent of digital TVs and set-top boxes, 95 per cent of smartphones and 10 per cent of mobile computers. As of 2014, ARM is the most widely used 32-bit instruction set architecture in terms of quantity produced



The ARM1 second processor for the BBC Micro (Courtesy: en.wikipedia.org)

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Millions of electronic gadgets around the world invade our daily life and we have become completely dependent on them for performing most of our work. A very nice example to justify this would be a smartphone that everyone is quickly adapting to, due to its varied features. These gadgets are the pleasing results of the ceaseless developments in the field of electronics. Most of these gadgets are fitted with embedded processors that not only occupy less space but also ensure that users get a smooth experience whilst using the device. The processor cores of these devices follow an architecture that helps them perform efficiently.

Nowadays, there are several embedded architectures in use such as ARM architecture developed by ARM Ltd, Atmel's AVR architecture, TI's MSP430 architecture and many more. However, the extensively used and most popular embedded architecture amongst many companies is the ARM Ltd's ARM architecture.

History of ARM

Advanced RISC machine (ARM) is the first reduced instruction set computer (RISC) processor for commercial use, which is currently being developed by ARM Holdings. The history of ARM processors dates back to 1983 in England when Acorn Computers Ltd officially launched an Acorn RISC Management project after being inspired to

design its own processor by Berkeley RISC, one of the high-impact projects under ARPA's (Advanced Research Projects Agency, now converted to DARPA) VLSI project, dealing with RISC-based microprocessor design led by David Patterson who coined the term 'RISC.' As the name suggests, it does not mean that the processors with less than 100 instructions are qualified to RISC category, but instead they should have an highly optimised instruction set.

ARM in the beginning was known as Acorn RISC machine. With VLSI Technology Inc. as its silicon partner, ARM came up with ARM1, the first ARM silicon on April 26, 1985, which was used as a second processor to the BBC Micro to develop the simulation software to finish work on the support chips (VIDC, IOC and MEMC) and to increase the operating speed of the CAD software used in development of ARM2. Apple, whilst developing an entirely new computing platform for its Newton, a personal digital assistant, found that only Acorn RISC machine was close to the requirements needed for implementation, but since ARM had no integral memory management unit, Apple collaborated with Acorn to develop ARM. The result of this collaboration was that both Acorn Group and Apple Computer, Inc., with 43 per cent share each, and VLSI Technology, Inc. as an investor, a separate company, ARM Ltd, was established in 1990. Also, the advanced research and development section of Acorn was employed here. After that time, ARM became the acronym for advanced RISC machine.